# Farzaneh Fayazbakhsh

Bioinformatics Researcher | Protein Structure, Genomics & Reproducible Computation

Stockholm, Sweden   fa.fayazbakhsh@gmail.com   079-346 08 88   fari-labwitch

## Personal Statement

Interdisciplinary bioinformatics and molecular-biology researcher with experience in sequencing analysis, genomics, structural-variant discovery and reproducible Python/R/HPC workflows. I bring foundational familiarity with AlphaFold and molecular-structure analysis alongside a strong translational-biology base, and am motivated to deepen my expertise in protein-structure informatics and evolution.

## Computational Biology & Bioinformatics Skills

**Programming & Pipeline Automation:** Python, R, UNIX/Linux Shell, Nextflow (pipeline development), Git, and GitHub for version control

**Genomics, High-Throughput Sequencing & Genetics:** Practical experience in NGS data analysis, Whole Genome Sequencing (WGS), RNA-Seq processing, GWAS, QTL analysis, gene mapping, structural variation, metagenomics, genome annotation, and phylogenomics (MEGA, Clustal Omega)

**Structural Biology, Modeling & Machine Learning:** Sequence analysis, primer design, AlphaFold (protein structure prediction), PyMOL (molecular visualization), and foundational familiarity with deep learning frameworks

**Infrastructure, HPC & DevOps:** Hands-on experience with High-Performance Computing (HPC) clusters, containerization (Docker, Docker Compose, Apptainer/Singularity), and implementation of FAIR Research Data Principles

## Bioinformatics & Computational Biology Projects

**Automated Raw Sequencing Data Quality Control Pipeline**

- Built an automated, parallelized Bash orchestration workflow designed for execution on High-Performance Computing (HPC) clusters via SLURM scheduling.
- Deployed FastQC and fastp dynamically inside isolated, version-controlled Singularity and Apptainer containers to automate raw sequence pairing, quality checking, and adapter trimming.
- Implemented multi-core processor allocation inside container runtimes to optimize throughput when executing string substitutions and batch file calculations.
- Orchestrated MultiQC container tasks to aggregate and output comprehensive data quality matrices, tracking processing adjustments across all sample cohorts simultaneously.

**Tumor Microenvironment Profiling: scRNA-Seq & Spatial Transcriptomics**

- Engineered end-to-end transcriptomic pipelines in R using Seurat and SCTransform to map high-dimensional gene expression across breast cancer tissue cohorts.
- Executed data pre-processing, Quality Control (QC), linear (PCA), and non-linear (t-SNE / UMAP) dimensionality reduction alongside graph-based clustering to resolve distinct cell populations and structural tissue features.
- Containerized both analysis environments using Docker and Docker Compose, ensuring full cross-platform compatibility and seamless runtime execution across varied infrastructure platforms.
- Managed pipeline documentation, software repositories, and strict workflow reproducibility utilizing Git and GitHub version control standards.

## Research Experience

**Karolinska Institute & SciLifeLab**, Master's Thesis Research — 2026–present

- Investigating replication-stress-induced genomic instability at common fragile sites using targeted long-read sequencing.
- Performing neuroblastoma cell culture, drug treatments, DNA extraction, and long-range PCR.
- Developed bioinformatics workflows using Python, Linux and HPC environments for long-read sequencing analysis.
- Performed structural variant discovery, breakpoint characterization and endogenous barcode identification.

**KTH Royal Institute of Technology & SciLifeLab**, Research Engineer 2023–2024
Sweden

- Independently developed Tumor-on-a-Chip models for in vitro anti-cancer drug screening.
- Optimized layer-by-layer cellulose nanofibril coatings to enhance cancer spheroid formation.
- Contributed to projects on Brain-on-a-Chip models for metabolic studies.
- Functionalized electrochemical sensors for ketone detection.

**Tehran University of Medical Sciences**, Master's Thesis Research 2018–2022
Iran

- Synthesized gold nanoparticles conjugated with polyphenols.
- Developed microfluidic devices for endothelial cell culture.
- Modeled early-stage endothelial dysfunction and conducted oxidative stress assays.
- Assisted in preparing a PET/PDMS/Collagen nanofibrous membrane for microfluidic cell culture.

**Gen-Iran Lab**, Pathology and Cytology Intern 2023
Tehran, Iran

- Hands-on experience in tissue preparation, fixation, embedding, and sectioning.
- Performed histological and immunohistochemical staining.

## Education

**M.Sc.** **Swedish University of Agricultural Sciences**, Bioinformatics 2025–present

- Highest grade (5/5) in Bioinformatics course
- Thesis: Exploring the Potential of Fragile Sites as Markers for Lineage Tracing

**KTH Royal Institute of Technology**, Single-Cell & Spatial Transcriptomics Data Analysis Course 2024

**M.Sc.** **Tehran University of Medical Sciences**, Medical Nanotechnology 2017–2022
Iran

- Awarded as the Top Master's Graduate (GPA: 4/4)
- Thesis: Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip

**B.Sc.** **Alzahra University**, Cellular and Molecular Biology (Biotechnology Specialization) 2012–2016
Iran

## Peer-Reviewed Publications

**Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip** 2023
**Farzaneh Fayazbakhsh**, Fatemeh Hataminia, Houra Mobaleghol Eslam, Mohammad Ajoudanian, Sharmin Kharrazi, Kazem Sharifi, Hossein Ghanbari
(Scientific Reports)

**Metabolic Assessment of iPSC-Derived Neurons Under Ketogenic and Normal Diets: Ketone Sensor Development and Comparative Analysis of Neuronal Metabolism** 2026
* Both authors contributed equally to this work (co-first authors).
Rohollah Nasiri*, **Farzaneh Fayazbakhsh***, Reyhaneh Sanei, Tingting Wu, Nayere Taebnia, Rouhollah Habibey, Omid Fotouhi, John Cognetti, Volker M. Lauschke, Aman Russom, Anna Herland
(CellPress)

**Assessing Layer-by-Layer Assembly of Cellulose Nanofibrils in Pancreatic Tumor Spheroid Formation** 2023
Negar Abbasi Aval, Ekeram Lahchaichi, Oana Tudoran, **Farzaneh Fayazbakhsh**, Rainer Heuchel, Matthias Löhr, Torbjörn Pettersson, Aman Russom
(Biomedicines)

**Characterization and Numerical Simulation of a New Microfluidic Device for Cell Investigations** 2024
H Meslam, F Hataminia, H Asadi-Saghandi, **F Fayazbakhsh**, N Tabatabaei, Hossein Ghanbari
(Frontiers in Lab on a Chip Technologies)

**Nanofibers in Respiratory Masks: An Alternative to Prevent Pathogen Transmission** 2022
Tabatabaei SN, Faridi-Majidi R, Boroumand S, Norouz F, Rahmani M, Rezaie F, **Fayazbakhsh F**, Faridi-Majidi R
(IEEE Transactions on NanoBioscience)

## Wet Lab & Experimental Expertise

**Cell Culture & Cancer Biology:** 2D & 3D cell culture, tumor spheroid formation and organoid culture, in vitro cancer biology drug screening

**Molecular Biology & Genomics Workflows:** DNA extraction, downstream cellular and molecular assessments, PCR / Real-Time PCR, agarose gel electrophoresis, Agilent TapeStation analysis, flow cytometry, fluorescent microscopy

**Microfluidics & Biosensors:** Organ-on-a-Chip modeling, PDMS-based microfluidic systems fabrication, surface and electrochemical sensor functionalization

**Histology & Pathology:** Histological sample preparation, H&E staining, immunohistochemistry (IHC)

**Nanotechnology & Nanomaterial Synthesis:** Green synthesis of gold nanoparticles, nanofiber electrospinning, cellulose nanofibrils (CNF) coating optimization

## Conference Presentations

**Spheroid Formation in Cellulose Nanofibrils-Coated Microfluidic Chips** 2024
µTAS 2024, Montreal, Canada
Poster presentation (first author)

**Design of a Microfluidic Electrochemical Ketone Sensor** 2023
Swedish Microfluidics in Life Science, KTH, Sweden
Oral presentation (co-author)

**Investigation of Effect of Different Diets on Brain Energy Metabolism Using Organ-on-a-Chip** 2023
MPS World Summit, Berlin, Germany
Poster presentation (co-author)

**Preparation of PET/PDMS/Collagen Nanofibrous Membrane Embedded in a Microfluidic Device for Organ-on-Chip Applications** 2022
The 3rd Conference on Nanofibers, Tehran University of Medical Sciences
Poster presentation (co-author)

## Professional & Personal Skills

**Bridging Wet & Dry Lab:** Confident working in both experimental labs and computational environments, making it easy to translate biological questions into data-driven answers.

**Independent Problem Solver:** Highly autonomous in research environments, with a proven ability to troubleshoot bottlenecks and quickly teach myself new tools to keep projects moving forward.

**Collaborative Team Player:** Experienced in international, interdisciplinary teams such as KTH, Karolinska Institute and SciLifeLab, with a strong focus on knowledge-sharing and helping colleagues.

**Clear Communicator & Fast Learner:** Skilled at explaining complex data clearly through peer-reviewed publications and conference presentations; highly adaptable and quick to pick up new scientific topics and skills.

## Teaching & Mentorship

**KTH Royal Institute of Technology**, Intern Supervisor
Sweden
- Guided an intern in laboratory techniques.

**Volunteer Service**, Science Teacher
- Taught science to underprivileged children.

## Awards & Honors

**Top Master's Graduate** 2023
Tehran University of Medical Sciences

**LUSH Prize Candidate** 2022
Shortlisted for the Young Researcher Award.

**Darolfonoon Student Award** 2020
For voluntary assistance in nanofiber mask production during the COVID-19 pandemic.

**Top Article Award** 2014
First National and Student Conference on Applied Biology, Alzahra University, Iran
Prize winner.

**Top 5% in Iran's Konkour for Master's Degree University Entrance** 2016
Iran
Ranked 37th in Iran's national Konkour for the Master's Degree University Entrance Exam.

## References

**Dr. Kasper Karlsson**
Department of Oncology-Pathology, Karolinska Institute
kasper.karlsson@ki.se

**Dr. Erik Bongcam Rudloff**
Department of Animal Biosciences, Swedish University of Agricultural Sciences
erik.bongcam@slu.se

**Dr. Aman Russom**
Nanobiotechnology Division, Protein Science Department, KTH Royal Institute of Technology
aman.russom@scilifelab.se

**Cheng-de (Ernest) Liu**
Department of Oncology-Pathology, Karolinska Institute
chengde.liu@ki.se

**Dr. Anna Herland**
Nanobiotechnology Division, Protein Science Department, KTH Royal Institute of Technology
aherland@kth.se

**Dr. Rohollah Nasiri**
Radiation Oncology Department, Stanford University
rhnasiri@stanford.edu